Chapter 5

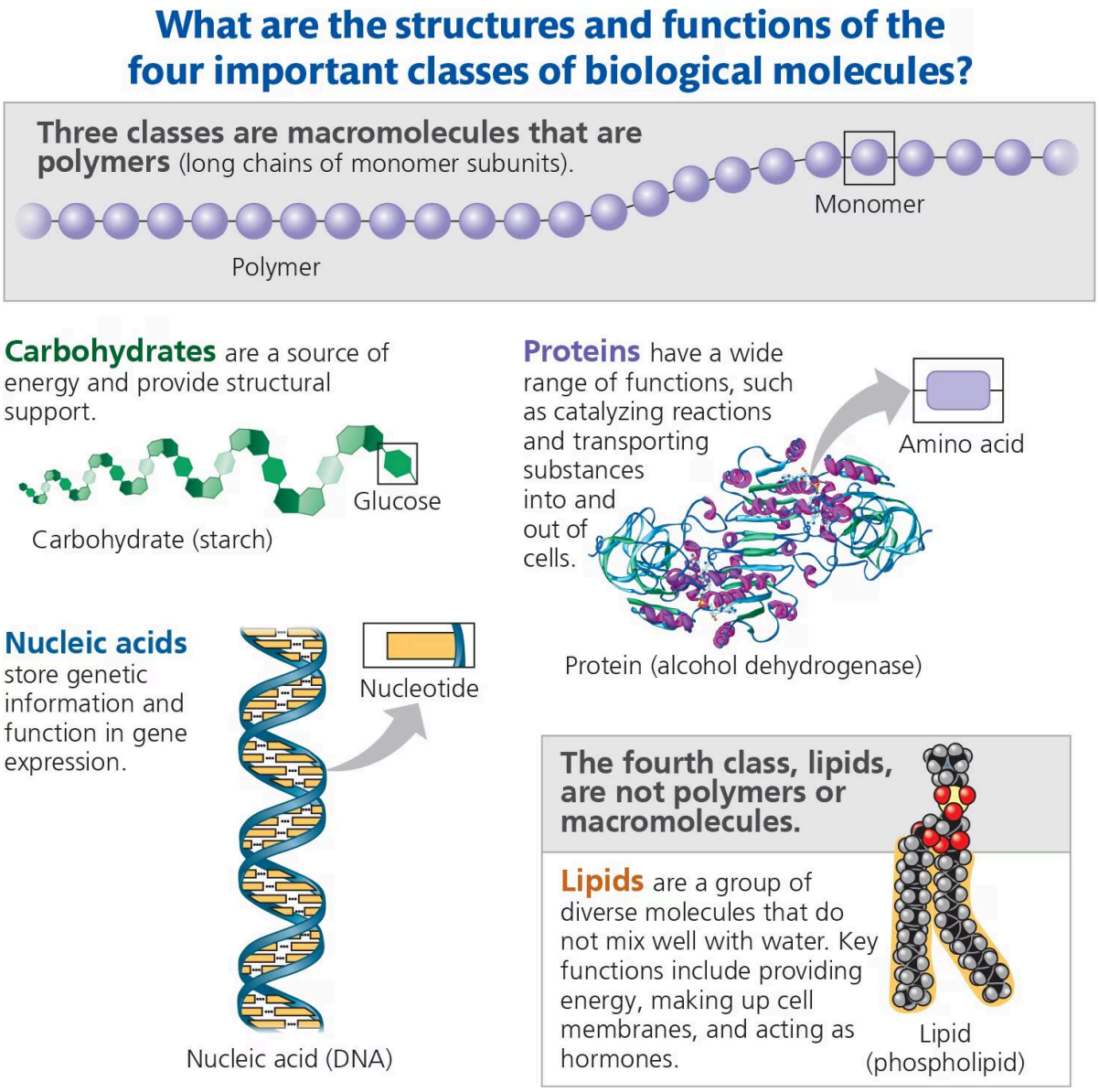
Campbell Biology, AP Edition, 12e

Chapter 5

The Structure and Function of Large Biological Molecules

Figure 5.1

Alcohol dehydrogenase, a protein that breaks down alcohol in the body, is shown here as a molecular model. The form of this protein that an individual possesses affects how well that person tolerates drinking alcohol. Proteins are one class of large molecules, or macromolecules.



Three classes are macromolecules that are polymers (long chains of monomer subunits). This is denoted by circles linked with lines to form a long chain, with one circle labeled monomer.

Carbohydrates are a source of energy and provide structural support. This is represented by a spiral made of hexagonal subunits, one of which is labeled glucose.

Proteins have a wide range of functions, such as catalyzing reactions and transporting substrates into and out of cells. A protein (alcohol dehydrogenase) is shown as a three-dimensional, coiled, ribbon-like structure where a small portion is expanded and labeled an amino acid.

Nucleic acids store genetic information and function in gene expression. An illustration of the DNA helix is shown, one portion of which is expanded and labeled a nucleotide.

The fourth class, lipids, are not polymers or macromolecules. Lipids are a group of diverse molecules that do not mix well with water. Key functions include providing energy, making up cell membranes, and acting as hormones. A phospholipid molecule is illustrated as a unit that has a head and two tails and made of globular subunits.

Key Concepts

- **5.1Macromolecules are polymers, built from monomers****
- **5.2Carbohydrates serve as fuel and building material****
- **5.3Lipids are a diverse group of hydrophobic molecules****
- **5.4Proteins include a diversity of structures, resulting in a wide range of functions****
- **5.5Nucleic acids store, transmit, and help express hereditary information****
- **5.6Genomics and proteomics have transformed biological inquiry and applications****

All life-forms on Earth share biochemical similarities **(BIG IDEA 1)** that include large, complex molecules synthesized from common building blocks **(BIG IDEA 4).**

Study Tip

Make a visual study guide: For each class of biological molecules, draw two examples and list their structural similarities and their functions.

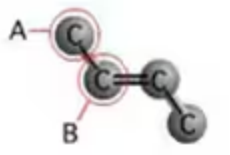
Important Biological Molecules	
Carbohydrates	Proteins
Nucleic acids	Lipids

Get Ready for This Chapter

In this chapter, you will learn about the structures and functions of the four classes of large biological molecules. Before beginning this chapter, you should be able to do the following:

- Explain the bonding behavior of a carbon atom (see Concept 4.2, section entitled “The Formation of Bonds with Carbon”).
- Identify the important biological functional groups (see Figure 4.9).
- Understand the chemical nature of functional groups (see Figure 4.9).

The following questions provide a quick check of your basic knowledge in these areas.



1. How many hydrogen atoms can be attached to carbon B?

1

That's correct.

Correct! If you had trouble with this question, review the following material: Each carbon atom can form four bonds. A double bond counts as two bonds, so carbon B has a total of three bonds so far, and it can form one more single bond with a hydrogen atom.

2

That's incorrect. Try again.

Incorrect. Review the following material: Each carbon atom can form four bonds. A double bond counts as two bonds, so carbon B has a total of three bonds so far, and it can form one more single bond with a hydrogen atom.

3

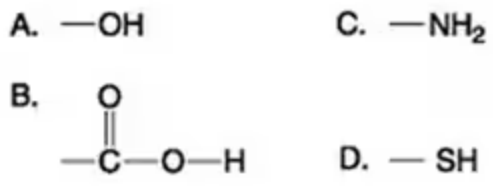
That's incorrect. Try again.

Incorrect. Review the following material: Each carbon atom can form four bonds. A double bond counts as two bonds, so carbon B has a total of three bonds so far, and it can form one more single bond with a hydrogen atom.

4

That's incorrect. Try again.

Incorrect. Review the following material: Each carbon atom can form four bonds. A double bond counts as two bonds, so carbon B has a total of three bonds so far, and it can form one more single bond with a hydrogen atom.



2. Which functional group(s) is (are) present in all amino acids?

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B and D
```

That's incorrect. Try again.

Incorrect. Review the following material: An amino acid has an amino group (-NH2) and a carboxyl group (-COOH).

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A and B
```

That's incorrect. Try again.

Incorrect. Review the following material: An amino acid has an amino group (-NH2) and a carboxyl group (-COOH).

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B and C
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That's correct.

Correct! If you had trouble with this question, review the following material: An amino acid has an amino group (-NH2) and a carboxyl group (-COOH).

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C only
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That's incorrect. Try again.

Incorrect. Review the following material: An amino acid has an amino group (-NH2) and a carboxyl group (-COOH).

3. Which functional group can act as an acid?

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amino
```

That's incorrect. Try again.

Incorrect. Review the following material: An acid can donate H+. The carboxyl group (-COOH) can donate H+ (forming -COO-) because the covalent bond between oxygen and hydrogen is so polar. It is the carboxyl group of amino acids that makes them acids.

```
carbonyl
```

That's incorrect. Try again.

Incorrect. Review the following material: An acid can donate H+. The carboxyl group (-COOH) can donate H+ (forming -COO-) because the covalent bond between oxygen and hydrogen is so polar. It is the carboxyl group of amino acids that makes them acids.

```
methyl
```

That's incorrect. Try again.

Incorrect. Review the following material: An acid can donate H+. The carboxyl group (-COOH) can donate H+ (forming -COO-) because the covalent bond between oxygen and hydrogen is so polar. It is the carboxyl group of amino acids that makes them acids.

```
carboxyl
```

That's correct.

Correct! If you had trouble with this question, review the following material: An acid can donate H+. The carboxyl group (-COOH) can donate H+ (forming -COO-) because the covalent bond between oxygen and hydrogen is so polar. It is the carboxyl group of amino acids that makes them acids.

4. Which functional group can act as a base?

```
amino
```

That's correct.

Correct! If you had trouble with this question, review the following material: A base can accept H+ from the surrounding solution. The amino group (-NH2) can accept H+ from the surrounding solution, forming -NH3+. Therefore, the amino group can act as a base.

```
carbonyl
```

That's incorrect. Try again.

Incorrect. Review the following material: A base can accept H+ from the surrounding solution. The amino group (-NH2) can accept H+ from the surrounding solution, forming -NH3+. Therefore, the amino group can act as a base.

```
carboxyl
```

That's incorrect. Try again.

Incorrect. Review the following material: A base can accept H+ from the surrounding solution. The amino group (-NH2) can accept H+ from the surrounding solution, forming -NH3+. Therefore, the amino group can act as a base.

```
        methyl
```

That's incorrect. Try again.

Incorrect. Review the following material: A base can accept H+ from the surrounding solution. The amino group (-NH2) can accept H+ from the surrounding solution, forming -NH3+. Therefore, the amino group can act as a base.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 5.1Explain the processes by which polymers are assembled and disassembled.
- 5.2Describe the structures and functions of simple and complex carbohydrates.
- 5.3Describe the structures of three kinds of lipids and explain their functions.
- 5.4List the main functions of proteins and describe the structures of amino acids and proteins.
- 5.5Compare and contrast the structures of DNA and RNA and their component nucleotides, and describe the functions of these polynucleotides.
- 5.6Explain how our ability to rapidly sequence DNA has affected biological inquiry and applications.